

PART 1 -- GENERAL

1.1 SCOPE OF WORK: This specification covers furnishing and installing twelve new submerged turbine mixer-aerator systems in the Aeration Basin 1 (East Campus).

1.2 CONTRACTOR SUBMITTALS: The Contractor shall submit shop drawings and other requirements per Section 01 33 00 including:

- a. Motor horsepower for maximum load condition at 5,100 elevation,
- b. Shaft sizes,
- c. Power factor at design load,
- d. Motor efficiency at design load,
- e. Efficiency at design load,
- g. Safety factor of drive system.
- f. Structural design and shop drawings for the walkway and platform.
- h. Aeration calculations for the Submerged Turbine Aerators.

PART 2 -- PRODUCTS

2.1 SUBMERGED TURBINE AERATORS: The units will provide aeration and mixing in each zone of aeration for a complete mix activated sludge treatment process. All aerators shall include the motor, drive mechanism, gearbox, support structure with vibration absorbers, impellers, shaft, bottom bearing guide, air spargers, OSHA required guards, and all other parts and fittings required to place the aerator in operation.

The turbine and air sparger assembly shall be compatible with the concrete structures shown on the drawings. The assemblies shall be suitable for this oxygen transfer at an elevation of 5,100 feet and a sludge temperature range of 40 to 100 degrees Fahrenheit. The new assemblies shall be adequate for creating a completely mixed flow regime in domestic wastewater. The dimensions of each zone of aeration and its respective maximum actual oxygen requirement are listed below:

Zone of Aeration	Dimensions (L x W x SWD)	Number of units	AOR (pounds oxygen per hour)
Pre-Anoxic 1	59.3' x 24.3' x 16'	2	16.0
Pre-Anoxic 2	59.3' x 24.3' x 16'	2	16.0
Pre-Anoxic 3	32' x 25.4' x 16'	1	9.0
Sludge Reaeration	32' x 25.4' x 16'	1	48.3
Aeration 1	51.5' x 32' x 16'	2	103.5
Aeration 2	51.5' x 43' x 16'	2	74.8
Post-Anoxic	51.5' x 38' x 16'	2	21.7

Submerged metal parts shall be 304 stainless steel; if welded, the metal shall be 304L.

The air sparger will be compatible with the turbine for providing adequate oxygen transfer. The air sparger shall connect to air piping by 6-inch Class 125 standard flanged end connections. Provide supports suitable for anchoring the air sparger to the concrete floor.

The aerator shall be designed to adequately mix and keep in suspension aerated solids within a zone of influence equal to the basin size shown on the Drawings.

The associated motors shall be 480-volt, 3-phase motor with motor protection (type IP 55 or IP 65), and thermal protection using PTC resistors. The motor HP shall not be greater than 12.5- or 20-horsepower (altitude corrected), as indicated on the Drawings. The motor will be driven using a constant speed motor.

AGMA gear calculations and bearing rating calculations are to be submitted and shall be equal to an AGMA Class I for the motor horsepower at design output speed and 80,000 hours of B-10 life at this rating.

The shaft bearing assembly shall be designed to operate at or below 0.7 of the natural frequency of the system.

The aerator turbine shaft shall be stainless-steel with flanged connection to the gearbox shaft and impeller.

Each agitator shall be furnished with a turbine-type impeller. The maximum impeller diameter shall be 96-inches. The mixer body will be composed of high-quality fiberglass reinforced plastic (FRP) or stainless steel. The bottom bearing guide shall be made from stainless fixed below the unit on the basin floor.

The Contractor shall furnish HyperClassic Mixer/Aerator units by INVENT or AER-GS mixer units by AquaTurbo.

PART 3 -- EXECUTION

3.1 GENERAL: Shipping, handling, storage, and installation shall be in conformance with the manufacturer's recommendations.

3.2 INSTALLATION AND START-UP SUPERVISION

- A. Services of the manufacturer's factory trained representative, who is specifically knowledgeable in the type of equipment specified herein, shall be provided during the equipment installation period.
- B. Upon completion of the installation, the services of the manufacturer's factory trained representative shall be provided at the project site for equipment and performance testing. During the start-up, the manufacturer's representative shall inspect all system components for proper connection and alignment and assist the Owner in placing the equipment in a proper operating condition.
- C. Contractor to coordinate with Manufacturer and Engineer for schedule.

3.3 OPERATOR TRAINING: Upon satisfactory completion of the start-up, the manufacturer's representative shall instruct the Owner's personnel in the proper operation and maintenance of the equipment.

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